Math 2203 | Final (Sample) | Print Name ______________________
Spring 2026 | Section __________ | KSU ID ______________________

Directions: Read all the questions carefully and answer them completely. No credit will be given for undocumented responses. **Show all your work to get full credit.** Give **exact answers** to questions unless explicitly requested otherwise.

1. (10 pts) Find the velocity vector $\mathbf{v}(t)$, given the acceleration vector $\mathbf{a}(t) = \langle 1, e^t, -2t+1\rangle$, and the initial velocity vector $\mathbf{v}(0) = \langle -2, 4, 3\rangle$.

2. (10 pts) Compute the first-order partial derivatives of $z = \frac{\sqrt{x^2+y^2}}{y+1}$.

   $\dfrac{\partial z}{\partial x} =$

   $\dfrac{\partial z}{\partial y} =$

3. (10 pts)

   (a) Find the equation of the tangent plane to the graph of $f(x, y) = xy^2 + x^3 - y$ at the point $(-1, 1)$.

   (b) Use it to estimate $f(-1.1, 1.1)$

4. (10 pts) Find the directional derivative of $f(x, y, z) = yz + x^3$ at the point $P = (-1, -5, -1)$ in the direction $\mathbf{v} = \langle 1, -2, 3\rangle$.

5. (10 pts) Evaluate the integral $\iint_{\mathcal{D}} e^{x+y}\, dA$ over the domain $\mathcal{D}$ defined by $x \geq 0$, $y \leq 0$, $y - x \geq -3$.

6. (10 pts)

   (a) Find the critical point(s) of the function $f(x, y) = x^3 - 3xy + y^3$.

   (b) Use the Second Derivative Test to determine the local minimum, local maximum, and saddle points.

7. (10 pts) Calculate the integral of $f(x, y) = y$ over the region $2 \leq x^2 + y^2 \leq 4$ by **changing to polar coordinates**.

8. (10 pts) Let $\mathbf{F} = \langle 3yz, 7xz, 11xy\rangle$. Calculate div($\mathbf{F}$) and curl($\mathbf{F}$).

9. (10 pts) Find the equation of the plane passing through the three points $P = (2, -1, 0)$, $Q = (1, 0, 1)$, and $R = (0, 1, -2)$:

   (a) Find the normal vector to the plane.

   (b) **Use point** P to set up the equation.

10. (10 pts) Find an arc length parametrization $\vec{r}_1(s)$ of the curve $\vec{r}(t) = \left\langle 2\sin(t), 2\cos(t), \sqrt{5}t\right\rangle$. Calculate the curvature function for $\vec{r}$.